

Article

Innovative Integration of Poetry and Visual Arts in Metaverse for Sustainable Education

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Abstract: The rapid advancement of digital technology has necessitated a reevaluation of traditional educational methodologies, particularly in literature and visual arts. This study investigates the application of metaverse technology to integrate contemporary poetry and visual arts, aiming to enhance university-level education. The purpose is to develop a convergent teaching method that leverages the immersive and interactive capabilities of the metaverse. The research involves a joint exhibition project with students from Sangmyung University and international participants, incorporating a metaverse-based educational program. A sample of 85 students participated in the program, and their experiences were evaluated through surveys and focus group interviews (FGIs). The findings reveal significant correlations between content satisfaction and method satisfaction, underscoring the importance of engaging and interactive methods. The study also identifies technical challenges and provides insights for optimizing digital platforms for educational purposes. The implications suggest that integrating metaverse technology in arts education can significantly enhance creativity, critical thinking, and interdisciplinary skills, offering a sustainable and innovative approach to modern education. Based on these implications, this paper proposes methods for incorporating the insights gained from case analyses and implications into the design of educational programs. It is anticipated that this approach will contribute to enhancing the quality of convergence education in higher education institutions. Furthermore, it is expected that this program will serve as a starting point for the systematic implementation of integrated education and the use of digital platforms, thereby helping to reduce disparities in integrated education between countries.

Keywords: teaching and learning method; convergence art education; metaverse education; the convergence education program of literature and design; education for the digital-native generation

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1. Introduction

In recent years, the rapid advancement of digital technology has necessitated a reevaluation of traditional educational methodologies, especially within literature and visual arts [1]. This study proposes an innovative approach by integrating contemporary poetry and visual arts within a metaverse environment to enhance university-level education. The objective is to create a convergent teaching and learning method that harnesses the immersive and interactive capabilities of the metaverse to foster creativity, critical thinking, and interdisciplinary skills among students. The integration of arts into education is well established for its ability to enhance esthetic sensitivity and creativity. The 2015 revised curriculum, announced by Korea's Ministry of Education as the nation's 10th curriculum, emphasizes esthetic sensitivity as a core competency across elementary-, middle-, and high school education. It defines this competency as the ability to discover and appreciate the meaning and value of life through empathetic understanding and cultural sensitivity. [2]. This aligns with the broader educational objective of fostering holistic development through interdisciplinary learning, which has been shown to be beneficial in various studies [3,4].

As students educated under the revised curriculum matriculate at universities, Korean institutions are exploring various methods to enhance esthetic sensitivity. Consequently, an increasing number of university curricula are being designed to improve cultural sensitivity through interdisciplinary arts education. This study seeks to examine effective and innovative educational methods that address these current needs.

However, despite the pressing need for a thorough consideration of effective strategies for designing teaching methods and educational programs that meet the educational needs of both educators and students in the classroom, the literature has yet to fully explore the practical application of digital environments, specifically the metaverse, in integrating arts education. Previous studies have primarily focused on traditional methods or isolated digital tools without fully exploring the potential of immersive technologies to create comprehensive learning experiences [5,6]. This study addresses this gap by examining how the metaverse can be used to integrate contemporary poetry and visual arts, providing a holistic and engaging educational experience for digital-native students. Moreover, this study underscores the necessity of method satisfaction in enhancing content satisfaction. The findings suggest that engaging, interactive methods are crucial for effective learning. By leveraging the metaverse's immersive capabilities, educators can create more engaging and effective learning experiences, thereby improving educational outcomes [7,8].

Metaverse technology, which defines a virtual space, can be effectively used in education to create an expanded learning environment. It offers authentic learning experiences, facilitates interactions beyond the constraints of time and space, and can enhance students' motivation and interest. This demonstrates its diverse potential applications in educational contexts [9].

The implications of this study extend beyond the immediate context, offering practical insights for educators and metaverse developers on optimizing digital platforms for educational purposes. By addressing both technical challenges and pedagogical strategies, this research contributes to the broader discourse on sustainable and innovative education, highlighting the potential of the metaverse as a transformative tool in the integration of digital technologies and interdisciplinary education.

Augmented reality (AR) enables a deeper understanding of complex concepts that are difficult to convey through text alone and allows learners to construct knowledge through personal experience. Additionally, life logging in metaverse education facilitates reflective learning based on related data, enabling improvements and promoting customized learning trajectories. Moreover, Mirror Worlds transcend spatial and physical limitations, efficiently extending the real world and enhancing educational outcomes through the elements of fun, play, and collective intelligence [9].

This study proposes a structured approach to integrating contemporary poetry and visual arts in university education through the metaverse. By conducting a joint exhibition project involving students from Sangmyung University and international participants, this research aims to test the effectiveness of metaverse-based education. The program includes creating visual poetry posters, which are then exhibited in a virtual space. This hands-on approach allows students to engage creatively with both literary and visual elements while leveraging digital tools to enhance their learning experience. The effectiveness of this approach is evaluated through surveys and focus group interviews (FGIs) with the participants, providing both qualitative and quantitative data to inform future educational strategies.

The findings from this study offer several contributions to the field of education. Firstly, it demonstrates the feasibility and benefits of using the metaverse to create immersive, interdisciplinary learning environments that combine poetry and visual arts. This aligns with the need for innovative educational strategies that can adapt to the changing technological landscape and the preferences of the digital-native generation [10]. Secondly, it highlights the importance of method satisfaction in enhancing content satisfaction, suggesting that the integration of engaging, interactive methods is crucial for effective learning [11]. Finally, the study provides practical insights for educators and metaverse de-

velopers on how to overcome technical challenges and optimize the use of digital platforms for educational purposes, thereby contributing to the broader discourse on sustainable and innovative education.

Furthermore, starting in 2025, many Korean universities will expand and implement the “admission without a declared major” system for a portion of their enrollment quotas. This initiative focuses on fostering interdisciplinary talents by allowing students to independently choose their courses and design their career paths after admission. While this approach expands students’ range of choices and autonomy, it also presents the challenge of swiftly narrowing the gaps between traditionally distinct academic disciplines. This paper aims to design a program that integrates art and literature, which have traditionally been taught separately, using digital media.

By exploring the convergence of contemporary poetry and visual arts within a meta-verse environment, this study not only addresses a critical gap in the literature but also offers a novel educational paradigm that can be adapted and expanded in various educational settings. The implications extend beyond the immediate context, providing a foundation for further research and development in the integration of digital technologies and interdisciplinary education.

2. Theoretical Background

Attempting the convergence of poetry and visual arts in education can provide new educational stimuli for both art/design and literature majors. It also gives students, who live in a multimedia era familiar with visual culture, the opportunity to creatively express modern experiences and cultivate the creativity and originality needed for the future society.

This is achieved through what Dewey referred to as “aesthetic experience”. Kim Hye-young explains that Dewey’s emphasis on esthetic experience through art aligns with M. Johnson’s assertion of “meaning-making”, viewing Dewey’s approach to art as addressing “the entire process of inquiry that creates meaning from the rich, deep content of our undivided bodily and mental experiences as ‘the problem of art’” [12]. Re-recognized changes and disruptions through esthetic experiences lead to contemplating new directions [13].

In the current educational context, where there is limited access to and experience with poetry, an arts-based approach as a radical pedagogy that diverges from traditional poetry education can foster interest in poetry and serve as a new educational practice that goes beyond text-based literacy [14].

This chapter introduces the unique situation of Korea, which may differ from other countries, in applying the convergence of poetry and visual arts and the metaverse.

Recently, Korean contemporary poetry has begun to actively combine poetry and imagery, showcasing poetic innovations using photography, artworks, icons, and typography. Prominent examples include the poet Lee So-ho, whose works, such as “Catcalling” [15] and “Unseemly and Imperfect Letters” [16], reveal visual cultural influences and extensively use typography and design for visual effects. Various visual effects are being experimented with, and many poems use visual techniques, such as arranging poems like artworks. For instance, “Unseemly and Imperfect Letters” [16] is presented as a virtual art museum, introducing the poems as works displayed in the exhibition hall [17]. Kim Soo-i notes that authors are now using images and videos in literature to express themselves, moving beyond text, which aligns with the current trends [18]. Notably, this approach resembles the visual art of letters attempted in art. Among the various methods of fusing literature and art, this study aimed to combine literature and design. The convergence of literature and design has a long history. In the 1960s, France’s Oulipo (Ouvroir de littérature potentielle [tr. Workshop of Potential Literature]) explored letters as play by deconstructing the existing constraints and applying mathematical and geometric methodologies. Additionally, in the 1980s, visual poetry emerged, which utilized metaphorical and implicit expressions as poetic language, creating various variations and avant-garde expressions.

The fusion of literature and design has been pursued in various forms in Korea as well. Starting with Yi Sang's "Three-Dimensional Angle Blueprint—Memorandum on Lines" (1931) [19], a representative work of modern visual poetry, experimental typography, has been significantly addressed in the design field.

In Korea, the fusion of literature and design has proceeded in stages as follows: First, during the early stage—from the 1950s to the 1970s—popular literature and design developed as separate fields. Literature primarily evolved while maintaining traditional forms, whereas design thrived in printed materials and advertising due to economic development, existing as distinct domains rather than integrated fields. From the late 1970s to the early 1990s, typography began to gain attention in Korea as body text fonts started being utilized. Typography, which considers the shape and arrangement of letters to create visual expressions, played a role in promoting the integration of literature and design. During this period, attempts were made to visually express literary texts through typography. Since the 2000s, the advancement of digital technology has further invigorated the fusion of literature and design. Works combining literary content with design elements have emerged across various digital platforms such as websites, apps, and social media. From the 2020s to the present, the fusion of literature and design has expanded with new attempts in various media, mainly through online platforms such as webtoons and web novels.

Consequently, the fusion of literature and design in Korea has undergone various changes over time, now presenting a diverse array of forms. Examples of the fusion of literature and design include the following:

The first example is the 2010 exhibition held at the Korean Design Foundation Gallery to commemorate the 100th anniversary of the birth of the poet [20]. Based on the inspiration provided by Yi Sang, who was the first poet and designer in Korea to visually interpret texts and attempt experimental approaches, 45 members of the Korean Typography Society participated, reinterpreting and visually expressing Yi Sang's poetry through various media such as objects, graphics, and photographs. This fusion of design and poetry explored new possibilities in creation and art, contributing to the re-examination and expansion of diversity.

The second example, "Typojanchi: Weightless Writing" in 2013 [21], combined digital media and typography, departing from traditional print to explore new scenes where the language of poetry and typography blended as images, undertaken by seven teams of young poets and designers. This work was exhibited on the Seoul Square Media Canvas. Such collaboration contributed to rethinking and advancing how digital media can expand and enhance the expression of poetry.

The fusion of literature and design has traditionally focused on print media. However, there is now a growing need to expand this fusion by implementing the interaction between a video film and a text-based poem through a programming language and realizing it in AR and other various media [22]. Technology can expand literature and design into diverse expressive means, providing experiences where audiences can directly participate in story creation. Moreover, this fusion offers opportunities to explore new creative areas beyond traditional boundaries. Utilizing augmented reality technology allows for transitions to different perspectives, moving beyond printed stories. Therefore, the fusion of literature and design should be explored through various media to discover new experiences and domains through technological advancement and creative attempts.

This study attempted the fusion of literature and design through a metaverse exhibition. The program conducted in this study is significant in that it merged collaborative education, which was previously addressed in a dichotomous manner as literature content and design expression, enabling individuals to independently produce and express content and sought to expand its scope by incorporating the metaverse.

Recently, many instances can be found where the metaverse is utilized in the fields of literature and design education.

First, examining the field of literature, universities are making various efforts to provide literary experiences through the metaverse. In 2022, Yonsei University's Humanities Convergence Education Center held a metaverse content contest for a literary museum, creating a Yonsei Literature Museum metaverse map and soliciting digital content on the poet Yun Dong-ju. The Busan University of Foreign Studies held an exhibition of the Vietnamese author Hoang Minh Tuong's "The Poet Crosses the River" in its Southeast Asian Literature Exhibition Hall on its own metaverse platform. In 2023, the Seoul Institute of the Arts developed an audiobook *Ollang House* on a metaverse platform. Soongsil University's Global Future Education Center is building a metaverse school to teach Korean language and culture to global students inside a metaverse space.

Some studies have also explored the use of the metaverse for literary education in individual classrooms. In 2023, Lee [23] conducted research on the use of the metaverse in literary education, reporting that college students fully immersed themselves in creative activities using the metaverse and recognized it as a new literary space through a picture book creation class. In a 2022 study by Kim Soo-yeon [24], students experienced "participant intimacy" by engaging as "participants" in classical literature through the production and exhibition of videos using Zepeto and Gather.Town, thus making the boundaries of classics more flexible.

Examples can also be found in the design field where the metaverse has been applied to classes. In 2022, Jo [25] designed and experimented with a class where middle school second-year students planned and hosted an exhibition using the Mozilla Hubs Spoke metaverse platform. Art classes using the metaverse are increasingly spreading in various secondary education courses, aiming to expand students' creativity and imagination beyond physical limitations. In a 2023 survey of 120 design majors at D University, Kang et al. [9] found that the metaverse was primarily used for gaming and chatting rather than as an educational medium. The survey results indicated that its use in design classes was minimal, and students preferred the metaverse to be utilized in studio classes focused on practical work rather than just information delivery or comprehension.

In 2023, Lee [23] designed a class in G University's spatial design course where students used the metaverse platform Zepeto to create ideal public spaces. Creative problem-solving methods that differed entirely from traditional approaches emerged, such as designing mobility experiences distinct from reality or presenting concepts that transcend time, free from physical spatial constraints. Despite the various experiments and discussions on using the metaverse, there is still a need for more systematic research and development of educational programs utilizing the metaverse in regular university design and literary education.

As Jenkins pointed out in "Convergence Culture", metaverse education should be designed with high flexibility, respecting individual learners' experiences and intentions without right or wrong answers. This can help overcome the limitations of traditional cultural and arts education, which often result in disparities in interest or outcomes due to a focus on skills or talents, instead of only traditional lecture-based education [26].

3. Methodology

3.1. Research Subjects

This study comprises two parts. First, a survey was conducted. After conducting a joint exhibition project with Sangmyung University students and international university participants, a survey was conducted on those who participated in the metaverse exhibition, and the results were analyzed. The second part is an FGI. In-depth group interviews were conducted with the students who created works combining poetry and design, and their responses were reviewed.

The extracurricular program incorporating the metaverse was held between 24 May and 29 October 2023. This program included a contest targeting international university students. The participants first received basic education on creating convergent artworks combining poetry and visual arts. The aforementioned visual poetry involves selecting

specific poems and arranging typography elements such as words and phrases to create visual images. In this program, visual poetry was created in the form of posters, focusing on maximizing design principles such as emphasis and color contrast for virtual space exhibitions in the metaverse. Specifically, in the process of visualizing the text, guidance was provided to avoid using excessively small text considering the medium's characteristics. The final outputs were to be submitted in A2 size, 150 dpi, and in jpg or pdf format.

Figure 1 shows a student's completed work. They utilized a sufficient text size to ensure its readability and added texture to make it more image-like than text-like. The work, which raised issues and awareness about environmental problems, effectively used strong color contrasts to highlight a socially critical perspective. This was a particularly attention-grabbing element in the metaverse exhibition.

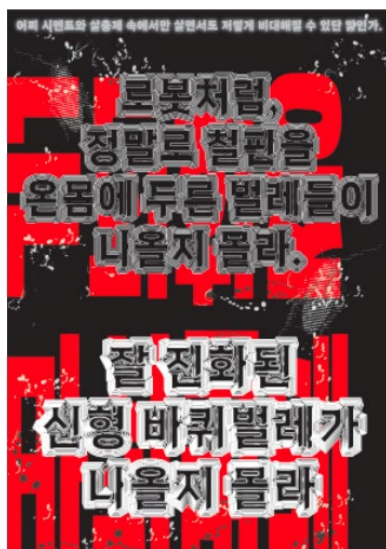


Figure 1. Work by student Jo Eun-young. Translation of Poem: Robots, like insects with metal plates covering their entire bodies, might emerge. New, well-refined cockroaches might appear.

Considering that the students might find the fusion of literature and visual arts unfamiliar, two preliminary special lectures were conducted (on 24 July 2023, practical training for creating exhibition works was conducted at Sangmyung University, and on 29 July, a real-time online special lecture via Zoom was held). Subsequently, the students' works were collected, and selected works were exhibited. Both online and offline exhibitions were held. An initial exhibition at Sangmyung University's Cheonan Campus Design Gallery took place from 29 to 31 August, followed by a public exhibition at the Cheonan Sinbu Cultural Hall (Exhibition Room 2) from 12 to 17 September. In addition, the works were uploaded to an online website, and a virtual exhibition was also conducted. Considering the characteristics of the digital-native students, digital technology was used for creation and a virtual reality (VR) (metaverse) exhibition was held. The metaverse exhibition platform used was Meum (<https://meum.me/>).

The actual appearance of the metaverse exhibition can be referenced in the image attached below (Figures 2–4). This image illustrates the layout of an exhibition space implemented within a metaverse environment, visually depicting how users interact and experience artwork in virtual reality. By examining such real-world implementations of metaverse exhibitions, we can explore the applicability of this technology and its potential future developments in a more concrete manner.



Figure 2. Entrance to the metaverse exhibition hall. (The Korean language in the image reads “2023 KF Public Diplomacy Project”).

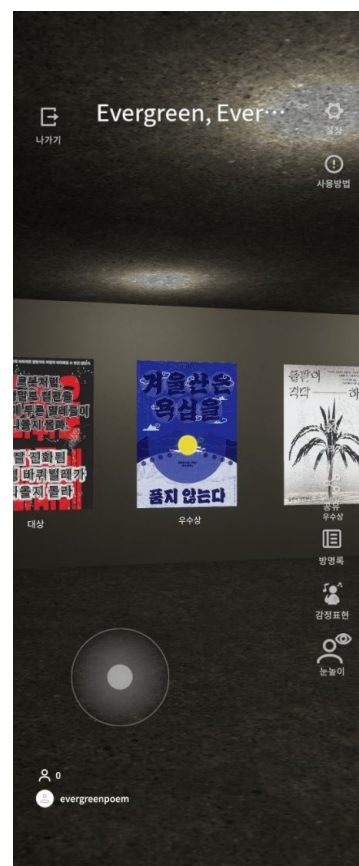


Figure 3. Scene from the metaverse exhibition (Korean on the screen is simply an instruction menu such as “move” “visit book” and “setting”).

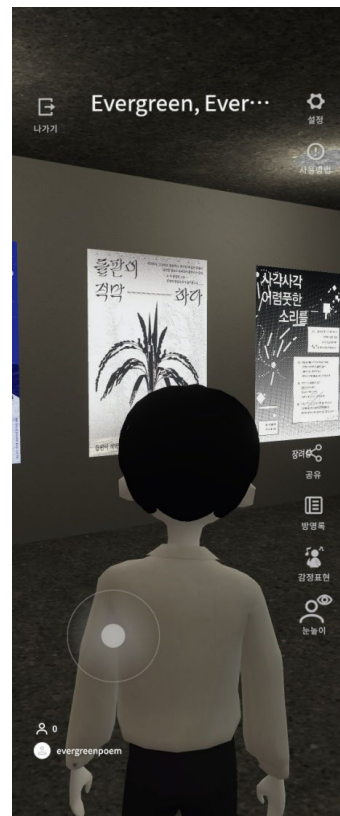


Figure 4. Avatar viewing the metaverse exhibition. (Korean on the screen is simply an instruction menu such as “move” “visit book” and “setting”).

Some works incorporated various media effects, such as video and music, providing multimedia expression for the students and a multimedia experience for the viewers.

Approximately 300 participants were recorded for the metaverse exhibition; however, it is presumed that the actual number is much higher due to the unregistered visitors in the metaverse exhibition hall guestbook, including internal personnel. With the offline exhibition held concurrently, the total number of program participants exceeded 500, but the exact number is not precise as it is believed to be higher than the recorded figures.

After the metaverse exhibition ended, a survey was conducted with the visitors. Its purpose was stated at the top of the questionnaire, informing the participants that the collected data would be used for research, and they were asked to fill it out if they agreed. As it was voluntary and anonymous participation, fewer responses were collected than the total number of visitors. The number of survey respondents was 85, all of whom were Sangmyung University students. The survey was conducted with the consent of the attendees at a metaverse exhibition, resulting in participation by 85 visitors who are all students in humanities or arts education. While the focus group interviews involved the students actively engaged in creative projects, the survey specifically targeted the metaverse exhibition visitors.

3.2. Research Methods

The survey was statistically analyzed using the statistical analysis software R 4.3.2. Basic statistical analysis (cross-analysis), correlation analysis, logistic regression analysis, and structural equation modeling analysis were conducted based on the survey responses to analyze the characteristics of the subjects and their responses to each question.

Method of analysis

- **Basic Statistical Analysis (Cross-Analysis):** Descriptive statistical analysis was conducted to examine the differences in the responses based on the participants' willingness to participate.
- **Correlation Analysis:** A correlation analysis was performed to investigate the degree of alignment in the respondents' answers to various questions.
- **Logistic Regression Analysis:** Logistic regression analysis was conducted with the willingness to participate as the dependent variable, identifying which questions significantly influenced the future willingness to participate.
- **Structural Equation Modeling (SEM):** Structural equation modeling was employed to group the items into factors and analyze the relationships. The results indicated that both the content and the methods of analysis significantly affected each other.

First, descriptive statistics were performed based on the participants' willingness to participate, and a correlation analysis was conducted to examine how the respondents' answers correlated with each other. Then, a logistic regression analysis was conducted—with "willingness to participate" as the dependent variable—to determine which questions influenced the future willingness to participate. A structural equation modeling analysis was used to group and analyze each question by factor.

The analysis results showed that the content and methods significantly influenced each other. The survey questions included questions about content (Q1_1. Usefulness of the program/event, Q1_2. Appropriateness of topic selection, Q1_3. Appropriate level of understanding, and Q1_4. Overall satisfaction with content), questions about methods (Q2_1. Interest in the method of combining contemporary poetry and visual arts, Q2_2. Appropriateness of the metaverse exhibition method, and Q2_3. Overall satisfaction with the method), questions about the topic (Q3_1. Usefulness of the topic, and Q3_2. Whether providing an immersive experience in introducing the topic was effective in artistically and esthetically experiencing the topic), questions about accessibility (Q4. Satisfaction with accessibility of the metaverse exhibition), questions about outcomes (Q5. Satisfaction with using works combining poetry and visual arts), questions about effectiveness (Q6_1. Whether the metaverse exhibition created an effective platform for collaboration and Q6_2. Effectiveness of the new way of appreciating poetry), and other questions (Q7_1. Whether they would participate in such exhibitions again and Q7_2. Whether they support such programs being planned and provided as educational programs). The survey was designed as the 7-point Likert scale.

The FGI was conducted in depth with four visual design major students who participated in the exhibition production. They had a sufficient understanding of basic typography, color mechanics, and multimedia. The questions were structured to cover considerations for visualization before the exhibition from the participant's perspective, the pros and cons and areas for the improvement of the metaverse exhibition from the visitor's perspective, and the necessary elements for poetry writing and design convergence education from the visual design major's perspective. The interview questionnaire and students' answers used in the FGI are attached to the Appendix A.

4. Analysis Results

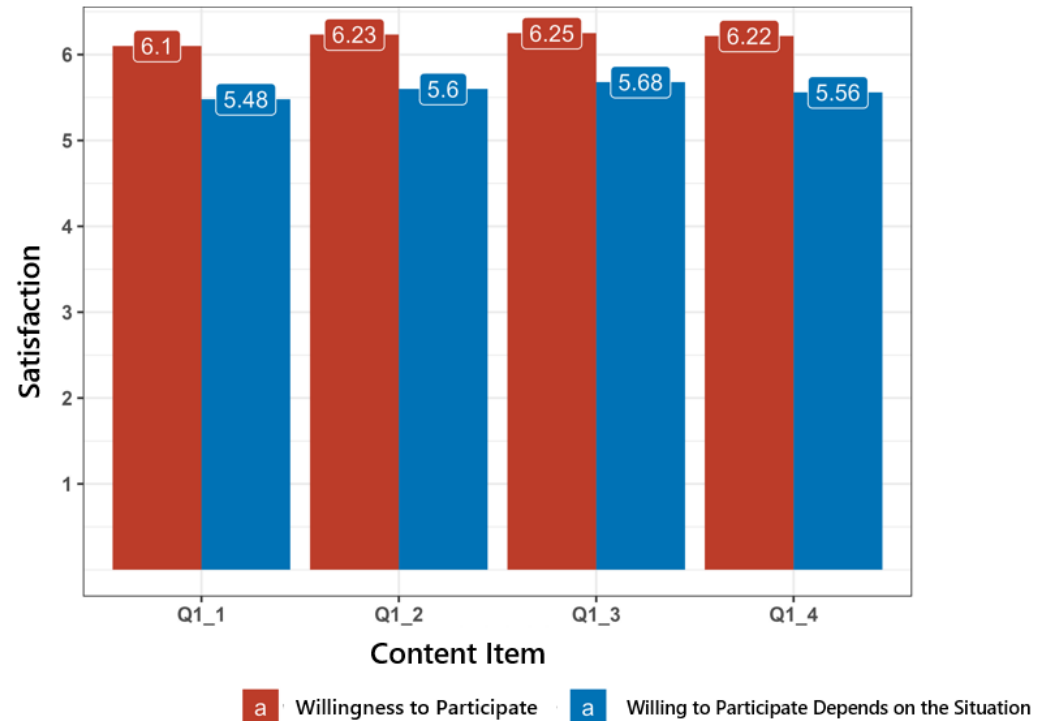
4.1. Basic Statistical Analysis Results

The responses to the question of whether the participants would participate again in such a program (Q7) were analyzed. Significant differences were found in content (Q1_2 and Q1_3) based on the participants' willingness to participate. The findings showed that the respondents who were willing to participate had overall higher satisfaction. These results are detailed in Table 1 and Figure 5, which analyzes responses to questions Q1_1 to Q1_4.

Table 1. Analysis of responses to questions (Q1_1 to Q1_4).

Question	Response	Willing to Participate (N = 60)	Depends on the Situation (N = 25)	p-Value
Q1_1	−4	4 (6.7%)	7 (28.0%)	0.051
	−5	14 (23.3%)	6 (24.0%)	
	−6	14 (23.3%)	5 (20.0%)	
	−7	28 (46.7%)	7 (28.0%)	
Q1_2	−4	5 (8.3%)	5 (20.0%)	0.018 **
	−5	6 (10.0%)	8 (32.0%)	
	−6	19 (31.7%)	4 (16.0%)	
	−7	30 (50.0%)	8 (32.0%)	
Q1_3	−4	4 (6.7%)	3 (12.0%)	0.014 **
	−5	9 (15.0%)	11 (44.0%)	
	−6	15 (25.0%)	2 (8.0%)	
	−7	32 (53.3%)	9 (36.0%)	
Q1_4	−3	0 (0.0%)	1 (4.0%)	0.112
	−4	4 (6.7%)	4 (16.0%)	
	−5	11 (18.3%)	8 (32.0%)	
	−6	13 (21.7%)	4 (16.0%)	
	−7	32 (53.3%)	8 (32.0%)	

N represents the sample size for each group. ** $p < 0.01$.

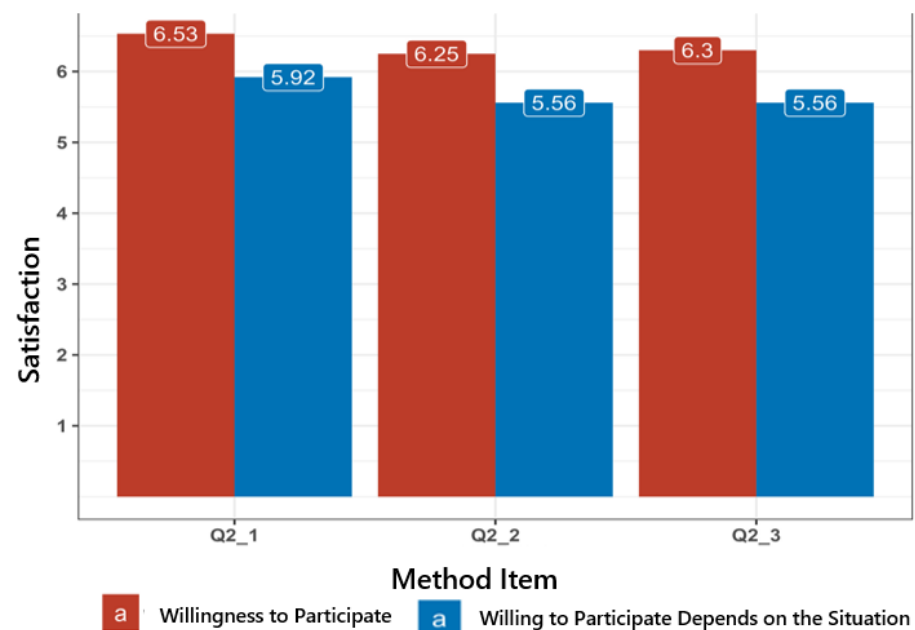
**Figure 5.** Basic statistical analysis of responses to questions (Q1_1 to Q1_4).

Significant differences in method (Q2_1 and Q2_2) were observed based on the willingness to participate. This confirms that the group with high satisfaction also has a high willingness to participate again. These results are detailed in Table 2 and Figure 6.

Table 2. Basic statistical analysis of responses to questions (Q2_1 to Q2_3).

Question	Response	Willing to Participate (N = 60)	Depends on the Situation (N = 25)	p-Value
Q2_1	−4	3 (5.0%)	3 (12.0%)	0.035 **
	−5	4 (6.7%)	6 (24.0%)	
	−6	11 (18.3%)	6 (24.0%)	
	−7	42 (70.0%)	10 (40.0%)	
Q2_2	−3	2 (3.3%)	0 (0.0%)	0.007 ***
	−4	2 (3.3%)	7 (28.0%)	
	−5	9 (15.0%)	6 (24.0%)	
	−6	13 (21.7%)	3 (12.0%)	
Q2_3	−7	34 (56.7%)	9 (36.0%)	0.054
	−3	0 (0.0%)	1 (4.0%)	
	−4	3 (5.0%)	4 (16.0%)	
	−5	9 (15.0%)	7 (28.0%)	
	−6	15 (25.0%)	6 (24.0%)	
	−7	33 (55.0%)	7 (28.0%)	

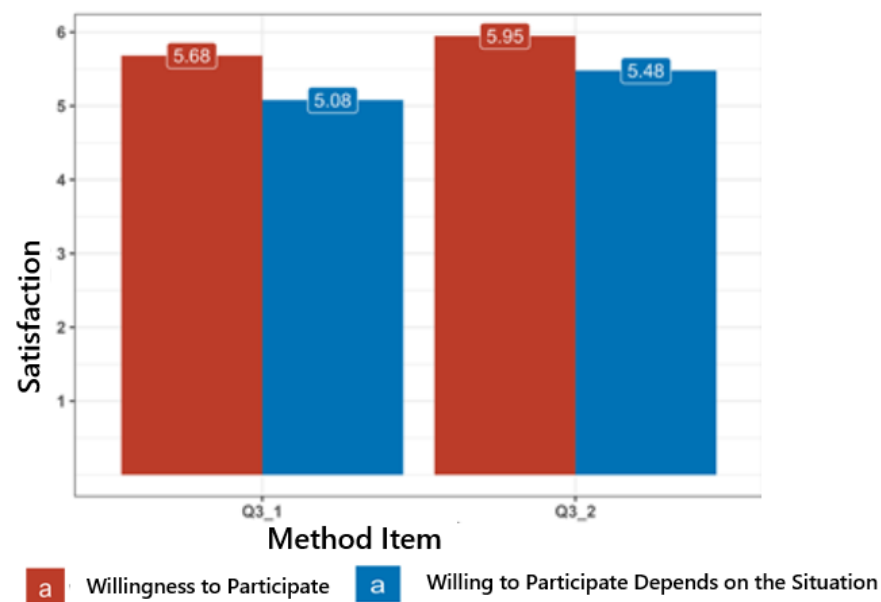
** p -value < 0.05. *** p -value < 0.01, indicating a statistically significant result.

**Figure 6.** Basic statistical analysis of responses to questions (Q2_1 to Q2_3).

In the following analysis, no significant differences in method (Q3_1 and Q3_2) were found based on the willingness to participate. Again, the respondents willing to participate had generally higher satisfaction. These results are detailed in Table 3 and Figure 7.

Table 3. Basic statistical analysis of responses to questions (Q3_1 to Q3_2).

Question	Response	Willing to Participate (N = 60)	Depends on the Situation (N = 25)	p-Value
Q3_1	−3	2 (3.3%)	3 (12.0%)	0.16
	−4	10 (16.7%)	6 (24.0%)	
	−5	13 (21.7%)	8 (32.0%)	
	−6	15 (25.0%)	2 (8.0%)	
	−7	20 (33.3%)	6 (24.0%)	
Q3_2	−3	1 (1.7%)	0 (0.0%)	0.187
	−4	7 (11.7%)	6 (24.0%)	
	−5	10 (16.7%)	8 (32.0%)	
	−6	18 (30.0%)	4 (16.0%)	
	−7	24 (40.0%)	7 (28.0%)	

**Figure 7.** Basic statistical analysis of responses to questions (Q3_1 to Q3_2).

4.2. Correlation Analysis

The correlation analysis was performed to examine the relationships between the questions, identifying which satisfaction items were related to each other. Detailed results can be seen in Table 4 and Figure 8 below.

As Table 4 and Figure 8 show, higher correlation coefficients indicate similar perceptions among the respondents, showing that those satisfied with the content were also highly satisfied with the method (1 and 3). That is, those satisfied with the metaverse exhibition method and the method of combining poetry and visual arts were also highly satisfied with the program and content (1, 2, 3, and 4). The topic questions had low correlation coefficients with the other questions, indicating that the exhibition topic did not affect the overall satisfaction. The accessibility questions had high correlation coefficients with content and effectiveness, showing that high satisfaction with the metaverse exhibition accessibility also led to high satisfaction with the content and educational effectiveness.

The outcome items generally had high correlation coefficients with all the items, especially with the effectiveness questions. The effectiveness questions had high correlation coefficients with the content (2, 3, and 4), method (1 and 3), and outcomes.

Table 4. Correlation analysis of responses to questions (Q1_1 to Q6_2). Means, standard deviations, and correlations with confidence intervals.

Variable	1	2	3	4	5	6	7	8	9	10	11	12
1. Q1_1												
2. Q1_2	0.77 ** [0.67, 0.85]											
3. Q1_3	0.69 ** [0.56, 0.79]	0.85 ** [0.78, 0.90]										
4. Q1_4	0.74 ** [0.62, 0.82]	0.77 ** [0.67, 0.85]	0.85 ** [0.78, 0.90]									
5. Q2_1	0.70 ** [0.57, 0.80]	0.78 ** [0.67, 0.85]	0.77 ** [0.66, 0.84]	0.82 ** [0.74, 0.88]								
6. Q2_2	0.57 ** [0.40, 0.69]	0.55 ** [0.38, 0.68]	0.60 ** [0.44, 0.72]	0.63 ** [0.49, 0.75]	0.56 ** [0.40, 0.69]							
7. Q2_3	0.70 ** [0.57, 0.79]	0.77 ** [0.66, 0.84]	0.85 ** [0.77, 0.90]	0.82 ** [0.74, 0.88]	0.82 ** [0.74, 0.88]	0.68 ** [0.54, 0.78]						
8. Q3_1	0.55 ** [0.38, 0.68]	0.64 ** [0.49, 0.75]	0.58 ** [0.42, 0.71]	0.64 ** [0.49, 0.75]	0.56 ** [0.40, 0.69]	0.67 ** [0.54, 0.78]	0.66 ** [0.52, 0.76]					
9. Q3_2	0.66 ** [0.52, 0.77]	0.73 ** [0.62, 0.82]	0.67 ** [0.53, 0.77]	0.62 ** [0.47, 0.74]	0.64 ** [0.50, 0.75]	0.67 ** [0.53, 0.77]	0.73 ** [0.61, 0.81]	0.72 ** [0.60, 0.81]				
10. Q4	0.55 ** [0.38, 0.68]	0.58 ** [0.42, 0.71]	0.60 ** [0.45, 0.72]	0.60 ** [0.45, 0.72]	0.70 ** [0.57, 0.80]	0.51 ** [0.33, 0.65]	0.66 ** [0.52, 0.77]	0.42 ** [0.23, 0.58]	0.52 ** [0.35, 0.66]			
11. Q5	0.68 ** [0.55, 0.78]	0.80 ** [0.70, 0.86]	0.74 ** [0.62, 0.82]	0.77 ** [0.66, 0.84]	0.82 ** [0.74, 0.88]	0.54 ** [0.37, 0.67]	0.78 ** [0.67, 0.85]	0.64 ** [0.50, 0.75]	0.71 ** [0.58, 0.80]	0.68 ** [0.55, 0.78]		
12. Q6_1	0.56 ** [0.40, 0.69]	0.73 ** [0.61, 0.81]	0.72 ** [0.60, 0.81]	0.68 ** [0.55, 0.78]	0.76 ** [0.66, 0.84]	0.47 ** [0.28, 0.62]	0.72 ** [0.59, 0.81]	0.56 ** [0.39, 0.69]	0.68 ** [0.55, 0.78]	0.70 ** [0.57, 0.79]	0.84 ** [0.76, 0.89]	
13. Q6_2	0.59 ** [0.43, 0.71]	0.71 ** [0.58, 0.80]	0.72 ** [0.59, 0.81]	0.70 ** [0.58, 0.80]	0.78 ** [0.67, 0.85]	0.49 ** [0.32, 0.64]	0.73 ** [0.62, 0.82]	0.50 ** [0.32, 0.64]	0.63 ** [0.48, 0.74]	0.70 ** [0.57, 0.79]	0.84 ** [0.76, 0.89]	0.85 ** [0.77, 0.90]

** indicates $p < 0.01$.

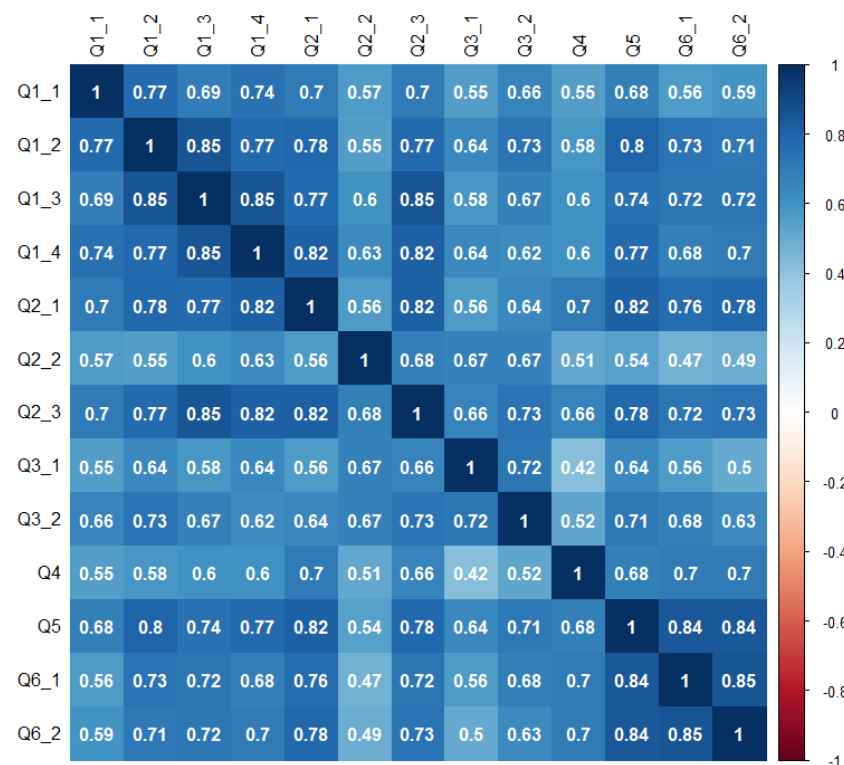


Figure 8. Correlation matrix.

4.3. Logistic Regression Analysis

A logistic regression analysis was conducted to determine which satisfaction factors influenced the future willingness to participate. As shown in Table 5, the analysis revealed that Q2_3 influenced participants' willingness to participate, indicating that higher satisfaction with Q2_3 increased their willingness to participate.

Table 5. Logistic regression analysis.

Variable	Estimate	Std. Error	p-Value
Intercept	−3.13	1.417	0.027
Q2_3	0.672	0.238	0.005

4.4. Structural Equation Modeling Analysis

Structural equation modeling was used to understand the influence relationships between the questions. The fit indices for the structural equation model are detailed in Table 6 below.

Table 6. The fit indices for the structural equation model.

Chisq	df	p-Value	GFI	RMSEA	CFI	TLI
98.856	38	0.000	0.846	0.134	0.940	0.913

- Chi-square (Chisq): This statistic assesses the difference between the model and the data. A smaller value is preferable. Here, the value is 95.856, and the p -value is 0.000, indicating significant differences.
- Goodness of Fit Index (GFI): This index represents model fit, where values above 0.90 are considered adequate. The model's GFI is 0.846, which is slightly below the threshold.

- Root Mean Square Error of Approximation (RMSEA): This index measures model error, with values below 0.05 considered adequate. The model's RMSEA is 0.134, indicating a lower fit.
- Comparative Fit Index (CFI): This index evaluates model fit, with values above 0.90 considered good. The model's CFI is 0.940, which is satisfactory.
- Tucker–Lewis Index (TLI): Another index for evaluating model fit, where values above 0.90 are preferable. The current TLI value is 0.913, which is relatively good. The indicators for evaluating convergent validity and reliability are presented in Table 7 below.

Table 7. The indicators for evaluating convergent validity and reliability.

	Reliability	Average Variance Extracted	Cronbach's Alpha
Q1	0.933	0.778	0.933
Q2	0.868	0.688	0.862
Q3	0.837	0.719	0.837
Q4	0.917	0.847	0.916

- Composite Reliability (CR): This evaluates the internal consistency of each construct. A CR value above 0.7 is considered to indicate good reliability. For Q1: 0.933, Q2: 0.868, Q3: 0.837, and Q6: 0.917—all indicate high reliability.
- Average Variance Extracted (AVE): This measures the explanatory power of each construct. An AVE value above 0.5 is considered adequate. For Q1: 0.778, Q2: 0.688, Q3: 0.719, and Q6: 0.847—all are considered adequate.
- Cronbach's Alpha: This index assesses internal consistency, where a value above 0.7 is viewed as indicative of high reliability. For Q1: 0.933, Q2: 0.862, Q3: 0.837, and Q6: 0.916—all demonstrate high reliability.

The findings showed that the method had the greatest impact on the content. As seen above, the higher the satisfaction with the method, the higher the satisfaction with the content, indicating that the method is crucial to participant satisfaction. The visualization results of the structural equation modeling (Figure 9) also show that Q1 and Q2 are connected by a thick line. These results are detailed in Table 8.

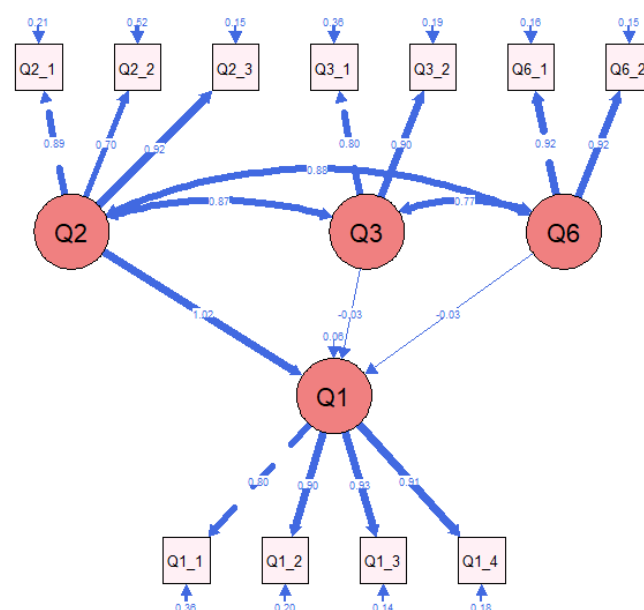


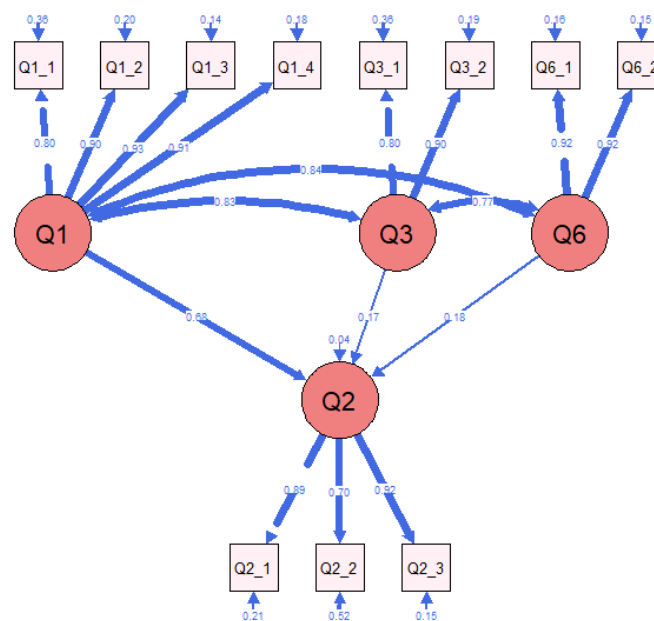
Figure 9. Visualization of structural equation modeling analysis of responses to questions (Q1_1 to Q6_2) (1).

Table 8. Structural equation modeling analysis (1).

Variable	Estimate	Std. Error	p-Value
Q2	1.050	0.263	<0.001 ***
Q3	−0.026	0.134	0.849
Q6	−0.030	0.148	0.842

*** p -value < 0.001, indicating a highly statistically significant result.

While content also impacts method, it is evident that the higher the satisfaction with the content, the higher the satisfaction with the method. Content and method are closely related, as confirmed by the thick line connecting Q2 and Q1 in the visualization results (Figure 10). Detailed results can be found in Table 9.

**Figure 10.** Visualization of structural equation modeling analysis of responses to questions (Q1_1 to Q6_2) (2).**Table 9.** Structural equation modeling analysis (2).

Variable	Estimate	Std. Error	p-Value
Q2	0.661	0.137	<0.001 ***
Q3	0.138	0.092	0.132
Q6	0.166	0.098	0.091

*** p -value < 0.001, indicating a highly statistically significant result.

For Q3, it was shown that the questions did not have significant relationships between the variables. Detailed results can be found in Table 10 and Figure 11.

Table 10. Structural equation modeling analysis (3).

Variable	Estimate	Std. Error	p-Value
Q1	−0.135	0.728	0.853
Q2	1.161	0.895	0.194
Q6	0.027	0.255	0.914

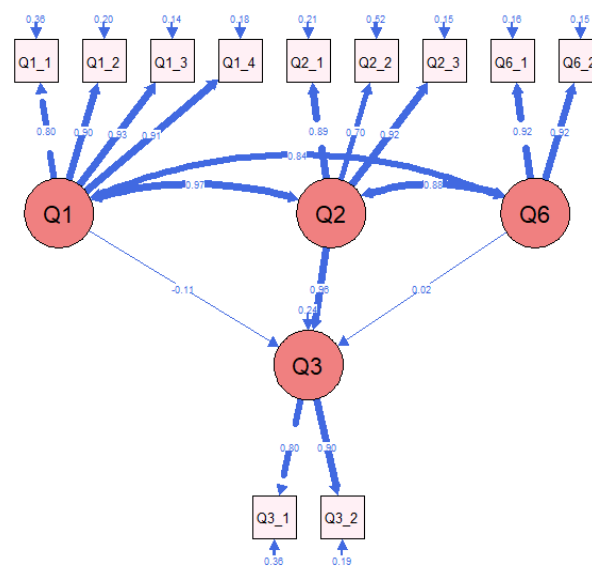


Figure 11. Visualization of structural equation modeling analysis of responses to questions (Q1_1 to Q6_2) (3).

It was also confirmed that the questions for Q6 did not have significant relationships between the variables, as shown by Figure 12.

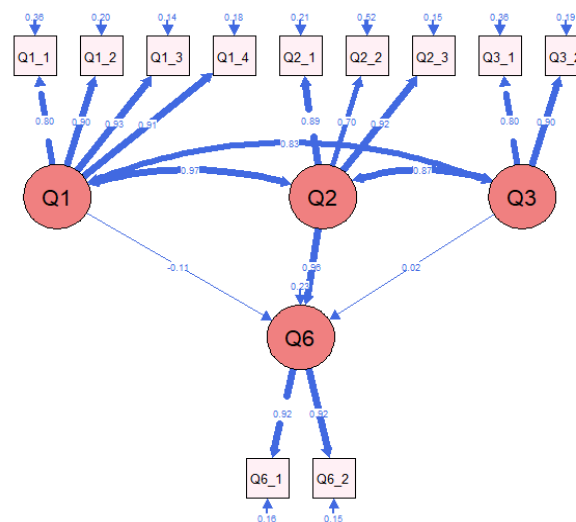


Figure 12. Visualization of structural equation modeling analysis of responses to questions (Q1_1 to Q6_2) (4).

4.5. FGI Results

The FGI with the visual design major students can be summarized as follows: First, in response to the question about specific considerations during the work process for participating in an exhibition using the metaverse, the participants focused on enhancing visibility and attention in the visualization work for digital media. There was a common opinion that in a digital environment, conditions such as clear color contrast and adjusting the size and shape of letters to increase readability are necessary to quickly read on small screens and guide the viewer's attention.

The second and third questions addressed the strengths and weaknesses of metaverse exhibitions and areas for improvement from the perspective of the exhibition visitors. In response to questions about what they liked and what they felt was lacking when viewing metaverse exhibitions, common opinions included the lack of time and space constraints

and the ability to engage actively, which provided a new experience. However, limitations such as difficulty in checking graphic details due to resolution loss and inconvenience in manipulation were also pointed out as the technical limitations of the digital environment. Regarding function, suggestions included addressing the limitations of digital exhibitions in terms of resolution and texture processing and maximizing the virtual experience of metaverse exhibitions by incorporating interactive elements and adding experiential content in the virtual world.

Finally, in response to a question regarding the essential education required in each field for designing future classes that integrate design and poetry writing, the respondents emphasized that basic idea generation and knowledge education are necessary for creative work in integrated poetry writing–design education. In terms of poetry writing, the respondents expressed the opinion that basic education on the fundamental concepts, structure, creative methods, and expressions of poetry is necessary and that on the design side, education is needed on analyzing the structure of poetry, visually translating it, and typography expression and layout. (Please refer to Appendix A for a detailed summary of the interview.)

5. Discussion

The analysis of the survey results indicates a significant correlation between content satisfaction and method satisfaction, underscoring the interconnectedness of these two aspects in educational programs. The respondents who were highly satisfied with the content also reported high satisfaction with the method, highlighting the fact that effective content delivery is heavily dependent on the chosen method. This finding is in line with previous research which suggests that the integration of content and method is crucial for enhancing educational outcomes [27,28].

The responses to the question about future participation (Q7) revealed significant differences in the content (Q1_2 and Q1_3) and method (Q2_1 and Q2_2) based on the participants' willingness to engage again. This suggests that satisfaction with the choice of topic and the level of understanding, as well as the presentation of the exhibition, plays a critical role in determining future engagement [29,30]. Similar findings have been reported in studies examining educational engagement, where content relevance and methodological appropriateness were key determinants of student participation [31,32].

Specifically, the results showed that the participants who expressed a high willingness to participate in future programs also reported higher satisfaction with the content and method. This aligns with the theory that well-structured content and engaging methodologies can foster a positive learning environment, thereby encouraging repeated participation [33]. The importance of method satisfaction, particularly in influencing future participation, cannot be overstated. This emphasizes the need for educators to carefully design and present their methods to enhance overall program satisfaction.

The correlation analysis further supports these findings, showing that satisfaction with the metaverse exhibition method and the method of combining poetry and visual arts were strongly related to overall program satisfaction. High correlation coefficients between these items suggest that the method of delivery significantly impacts content reception. This reinforces the idea that innovative educational methods, such as the use of the metaverse, can effectively enhance learning experiences and satisfaction [7,34,35].

The logistic regression analysis identified Q2_3 (overall satisfaction with the presentation) as a significant predictor of the willingness to participate in future programs. This highlights that the quality of the presentation is a crucial factor in maintaining student interest and engagement. The structural equation modeling analysis also showed that the method had the greatest impact on content satisfaction, illustrating that the way content is delivered is paramount to how it is received and perceived by students.

The FGI results provided additional insights into the students' perspectives on the metaverse exhibition. The participants emphasized the importance of clear color contrast, appropriate text size, and the overall readability of digital works, which are critical in

a digital environment where content is often viewed on small screens. This aligns with the findings from previous research on digital media, which stress the need for design considerations to enhance user experience in digital platforms [36].

The students also highlighted the strengths of metaverse exhibitions, such as the lack of time and space constraints and the ability to engage actively, which offered a new and enriching experience. However, they also pointed out the limitations related to resolution loss and manipulation inconvenience. These technical limitations suggest that while the metaverse offers novel opportunities for educational exhibitions, there are still challenges that need to be addressed to fully leverage its potential [37,38].

From an educational perspective, the respondents stressed the necessity of basic idea generation and knowledge education for creative work in integrated poetry writing–design education. This highlights the importance of foundational education in both fields to enable effective convergence. Previous studies have similarly emphasized the need for interdisciplinary education to foster creativity and innovation [3,4].

In summary, the results of this study underscore the critical role of method satisfaction in enhancing content satisfaction and overall program engagement. The findings suggest that innovative methods such as metaverse exhibitions can significantly enhance educational experiences by providing immersive and interactive learning environments.

6. Conclusions

6.1. Theoretical Contribution

This study offers significant theoretical contributions to the fields of art education and digital media integration. By examining the convergence of poetry and visual arts within the context of the metaverse, it provides fresh insights into how these disciplines can be synergistically combined to enhance educational outcomes. The innovative use of the metaverse as a platform for educational exhibitions addresses a gap in the literature regarding the practical application of digital environments in arts education.

One of the major theoretical contributions of this study is its exploration of the relationship between content and method satisfaction. While previous research has acknowledged the importance of engaging content and effective teaching methods [39,40], this study delves deeper into how these elements interrelate specifically within the context of a digital exhibition. The finding that method satisfaction significantly influences content satisfaction and overall engagement suggests a more intricate interplay between these factors than previously identified. This insight challenges the traditional approach of treating content and method as separate entities, instead proposing that their integration is essential for maximizing educational impact. Scholars can build on this by investigating other contexts where content–method integration might yield similar benefits.

Moreover, this study highlights the unique affordances and challenges of using the metaverse for educational purposes. Previous studies have discussed the potential of digital media to enhance learning experiences [41,42], but few have examined the specific role of the metaverse in this context. By identifying both the strengths, such as enhanced engagement and accessibility, and the limitations, such as technical difficulties and resolution issues, this research provides a comprehensive understanding of the metaverse’s practical applications. This dual perspective is crucial for developing more effective digital learning environments. Future research should explore how to mitigate the identified challenges, potentially by incorporating advanced digital tools and improving user interface design.

Chickering and Gamson proposed seven principles for effective undergraduate education practices designed to positively influence the student experience. These principles include enhancing interactions between faculty and students, fostering cooperative and reciprocal learning among students, encouraging active learning, providing prompt feedback, ensuring effective time management, setting high expectations, and cultivating respect for diversity [43]. Our study aims to implement these principles while exploring methods to further amplify their effectiveness.

Specifically, education is delivered on the metaverse platform, which transcends conventional spatial and temporal limitations, enabling “prompt feedback and effective time management” through technological utilization. The metaverse also facilitates increased interactions with faculty and peers. Moreover, the collaborative process of designing and preparing a metaverse exhibition serves as an educational strategy that promotes cooperative learning.

The creation and design process of a metaverse exhibition is inherently collaborative, with interactions occurring frequently. Visitors can freely leave feedback and comments, enabling them to express their views to the artists and assume an active role. Additionally, while viewing artworks in the metaverse exhibition space, visitors can also connect and communicate with other attendees.

Previous studies have not thoroughly examined how to integrate practical approaches to teaching poetry and design through technology. Our study addresses this gap by proposing a sophisticated academic framework that leverages the unique capabilities of the metaverse to enhance educational outcomes.

Additionally, this study offers valuable implications for scholars interested in interdisciplinary education. The emphasis on the need for foundational education in both poetry and design before attempting to integrate these fields is a critical insight. This study demonstrates that a deep understanding of both disciplines is necessary for successful convergence, suggesting that interdisciplinary programs should include comprehensive training in each field. This approach can help educators design more effective curricula that prepare students for the complexities of integrating diverse artistic disciplines. Scholars should further investigate how different combinations of disciplines can be effectively taught and what foundational knowledge is essential for each pairing.

As demonstrated by Simmons, the current expansion of undeclared and integrated majors, along with the increase in interdisciplinary studies students, signals an impending diversification and transformation within educational frameworks [44]. This highlights the critical importance of “active student engagement” and the provision of diverse engagement opportunities, especially for interdisciplinary studies students. Our research incorporates metaverse technology to enhance student participation and stimulate interdisciplinary studies. By leveraging digital technologies, we facilitate efficient time management and prompt feedback, offer immersive and diverse experiences, and enhance participation and communication. Therefore, our study not only addresses the expansion of undeclared major selection processes in Korea but also provides practical perspectives for designing educational programs for interdisciplinary studies students at international universities.

In summary, this study provides a detailed examination of the integration of poetry and visual arts in a digital environment, offering new perspectives on content–method interdependence, the practical use of the metaverse in education, and the requirements for successful interdisciplinary education. These contributions not only fill the gaps in the existing literature but also pave the way for future research to build on these findings, enhancing our understanding of how to effectively combine different forms of art and digital technology in educational settings.

6.2. Practical Implication

This study offers valuable practical implications for educators, metaverse developers, and immersive technology engineers. By exploring the integration of poetry and visual arts in the context of a metaverse exhibition, it provides a framework for utilizing digital platforms to enhance educational experiences and engagement.

For educators, particularly those involved in interdisciplinary and arts education, this study underscores the importance of integrating content and method effectively. Educators should design curricula that combine elements of poetry and visual arts, using digital tools to create immersive learning experiences. For instance, teachers can incorporate projects where students create visual poetry using digital design software such as adobe illustrator, photoshop and then display their work in a virtual exhibition space. This approach not

only engages students creatively but also develops their technical skills, preparing them for a digital future. Research highlights the effectiveness of such integrative methods in improving student engagement and learning outcomes [45].

Metaverse developers and immersive technology engineers can draw practical insights from this study's findings on the strengths and limitations of using the metaverse for educational purposes. Developers should focus on enhancing the technical aspects of virtual environments, such as improving resolution and user interface, to address the challenges identified by the participants. Incorporating high-resolution graphics and intuitive navigation tools can significantly enhance the user experience, making virtual exhibitions more accessible and enjoyable [46]. Moreover, adding interactive features like virtual tours or real-time collaboration tools can further enrich the educational potential of metaverse platforms [47,48].

The study also offers practical implications for sustainable education. By leveraging the metaverse, educational institutions can reduce their environmental footprint. The traditional physical exhibitions require significant resources for transportation, setup, and maintenance. In contrast, virtual exhibitions eliminate these needs, promoting a more sustainable approach to education. Schools and universities can adopt this model to host virtual art exhibitions, science fairs, and other events, thereby minimizing their carbon footprint while still providing enriching educational experiences.

Additionally, the integration of digital media in education, as suggested by this study, can bridge gaps in accessibility. Students from diverse geographical locations can participate in metaverse-based educational programs without the need for physical travel, making high-quality education more accessible to a broader audience. This is particularly beneficial for students in remote or underserved areas who may not have access to extensive educational resources. Schools can implement virtual exchange programs where students collaborate on projects and share their work in a virtual space, fostering global connections and cultural exchange [49].

To further support these practical applications, educational institutions should invest in training teachers to effectively use digital tools and platforms. Providing professional development opportunities that focus on the integration of digital media into curricula will equip teachers with the skills needed to create engaging and interactive learning experiences. Workshops and courses on digital design, virtual reality, and interdisciplinary teaching methods can help teachers stay abreast of technological advancements and apply them in their classrooms.

In conclusion, the findings from this study suggest practical steps for various stakeholders to enhance educational experiences through the integration of poetry, visual arts, and digital technology. By addressing technical challenges, promoting sustainable practices, and increasing accessibility, practitioners can leverage the metaverse to create more engaging and impactful educational programs.

This paper explores the design of a program that integrates the traditionally separate disciplines of arts and literature using digital media. The study also holds significance in exploring the potential of interdisciplinary research to lower the barriers between different majors for students who will enroll in universities under the “undeclared major, interdisciplinary studies” system in the future.

Considering that students, termed as digital natives, may possess a more profound understanding of digital environments than their instructors, it is imperative to acknowledge their unique attributes and ensure their perspectives are effectively integrated through communicative practices. Consequently, educational frameworks should be deliberately structured around a student-centric paradigm, promoting proactive and dynamic engagement from students in their educational journey.

This study addresses the gap in the existing research by exploring the practical application of digital environments, specifically the metaverse, in integrating arts education. It is anticipated that the findings will provide practical assistance to those designing inter-

disciplinary arts education programs that incorporate metaverse technology to integrate poetry and design in the future.

6.3. Limitations and Future Research Directions

This study aimed to develop a framework for exploring the potential of integrating interdisciplinary approaches before applying them to regular classes. As such, it has certain limitations, including a relatively small sample size and a focus on a specific educational context. Additionally, there is a need to address the technical issues encountered in the metaverse environment. These limitations lead to several recommendations for future research aimed at generalization. First, it is necessary to validate the findings with a larger and more diverse sample of students. This would allow for a better understanding of the long-term educational impacts of integrated learning activities facilitated by the “undeclared major, interdisciplinary studies” system and determine whether students are positively influenced by this program. Second, technical improvements must be made. The technical limitations identified through interviews—such as resolution problems, user interface constraints, and the limited use of interactive elements—need to be addressed. Furthermore, investigating advanced technological solutions to enhance the metaverse experience and conducting longitudinal studies to assess long-term educational impacts would be beneficial. Third, educators need to establish guidelines for essential foundational education in each field. For instance, in poetry writing, basic education on the fundamental concepts, structure, creative methods, and modes of expression is necessary. In design, there should be preliminary learning on analyzing poetic structure and translating it into visual language, as well as foundational education on layout and typography specific to digital media, which differs from analog media.

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Institutional Review Board Statement: This paper does not require approval from the ethics committee or the institutional review committee. We used an online non-face-to-face survey method using Google questionnaires, and all interviewees have agreed and the interviewees’ consent was obtained. In addition, there is no problem with research ethics because the survey notified the research utilization in advance and only those who wished participated anonymously.

Informed Consent Statement: Informed consent was obtained from all the subjects involved in the study.

Data Availability Statement: The data presented in this study are available on request from the corresponding author. The data are not publicly available due to privacy.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

Interview Questions

1. Pre-consideration

Were there any specific points to consider when participating in an exhibition using the metaverse?

2. The positive aspects and the shortcomings

What were the positive aspects and the shortcomings of the metaverse exhibition as you experienced it?

3. Improvement points

If you were to participate in a metaverse exhibition in the future, what improvements would you like to see? (e.g., features, planning, etc.).

4. Suggestions for Necessary Education

If there were a class that combines design and poetry writing, what kind of education do you think would be necessary in each field?

Interview Answers (summary)

Question 1

A: When creating graphic posters for use in the metaverse, I tried to design graphics suitable for a digital environment rather than on paper. It was important to make the colors vibrant by using high contrast colors and to enhance the readability of the titles. The layout and typography were kept simple and clean to ensure that the text was easy to read in a virtual space and that the main message could be quickly conveyed.

B: Because it was a metaverse exhibition, I minimized the use of various colors and ensured a strong contrast between the texts so that the posters could be easily seen in various environments.

C: Since digital works can be viewed from various mediums and angles, I applied designs that improve visibility and legibility. Also, to compensate for the lack of immediacy compared to offline exhibitions, I used bold text.

D: Since the exhibition is viewed in a digital environment, it was designed to maintain visibility without losing resolution. Ensuring viewers could focus on the work was important in the limited digital space. I worked with a certain degree of freedom regarding colors.

Question 2

Note: The second and third questions were about the pros and cons of metaverse exhibitions from the viewer's perspective and what could be improved in the future. A common opinion was that it provided a new experience by allowing active viewing without the constraints of time and space. However, technical limitations of the digital environment, such as difficulty in checking graphic details due to lower resolution and inconvenience in navigation, were also pointed out.

Interviewee A: It was nice to be able to easily view all the works through the metaverse exhibition. However, the detailed elements of the graphic posters were not clearly visible. In particular, small text or intricate graphic elements were not easily seen due to image quality issues.

Interviewee B: Viewing the exhibition through the metaverse is somewhat different, but it's fun and new, like a game. However, when moving large-sized posters into the metaverse environment, I felt that it was harder to focus on the viewing experience, and the posters' impact was reduced.

Interviewee C: The biggest advantage of a metaverse exhibition is that you can view it online. However, I was disappointed with the immersive space implementation and digital resolution.

Interviewee D: It was good that the exhibition could be easily viewed on the web and that there was an ability to focus on the work. However, the smoothness of navigation and some alien elements in the spatial representation were disappointing.

Question 3

Note: Functionally, there needs to be an improvement in the resolution and texture processing, which are limitations of digital exhibitions. There were also suggestions to maximize the virtual experience of metaverse exhibitions by incorporating interactive elements and adding experiential content to the virtual world.

Interviewee A: In future metaverse exhibitions, resolution and texture processing are needed to ensure graphics are always clear. It would also be good to add interactive elements so that viewers can better understand and enjoy the graphic posters.

Interviewee B: It might be beneficial to add interactions such as mouse clicks or hover effects to help focus the viewing of the posters. Creating web posters that only operate on the web could also be effective.

Interviewee C: It would be good to plan content that differentiates from other companies' metaverse showrooms. Offering experiential content could eliminate the boredom felt online and maximize the effectiveness of metaverse exhibitions.

Interviewee D: It's necessary to address technical issues when experiencing 3D spaces and to add better spatial representation or functionality. If there were unique experiential events that could only be experienced on the web, it could improve access and engagement with metaverse exhibitions.

Question 4

Note: Visual design majors emphasized that basic creative thinking methods and knowledge education are necessary for combining poetry writing and design. In terms of poetry writing, there is a need for basic education on the fundamental concepts, structure, creation methods, and expression techniques of poetry. On the design side, education that focuses on analyzing the structure of poetry and translating it visually, as well as on typographic expression and layout, was suggested.

Interviewee A: In an integrated education of design and poetry writing, the layout is crucial for effectively conveying the message in graphic poster design. I think it's important to learn techniques to visually express the emotions of poetry through the diversity of poetic expression, rhythm, and meter.

Interviewee B: It's important to learn the basic concepts of poetry writing. There should be education on exploring the structure of poetry in a design context and expressing the rhythm of poetry through text arrangement.

Interviewee C: Poetry writing classes should include education on the basic elements, creation methods, and various expression techniques of poetry. By working on expressing each poem in graphic design, one can learn how to combine the basic principles of design with typographic elements.

Interviewee D: In both fields, basic creative thinking methods and foundational knowledge needed for creation should be taught. In poetry writing, there should be education on the unity of the overall flow of the poem, and in design, education on using typography to letter and arrange poetry to complete a design is necessary.

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